

Cooking Times Reference

OptiMeal / Empyria - input for the compatibility validator - v0.1 draft

This is a draft written by Claude, not an authoritative source. Every number below is a starting point for Harris to correct. Nothing here is validated until it has been reviewed on paper, corrected, and signed. The protein section in particular carries safety weight and must not be treated as checked until it has been.

What this table is for

The compatibility validator answers one question: *can these ingredients go in the pan at the same time, cut this way?* It does not need exact minutes. It needs to know whether two things finish at roughly the same moment. So every ingredient resolves to a band.

Band	Time to done	Meaning
B1	0-2 min	Flash, wilt, or finish only
B2	3-5 min	Quick
B3	6-9 min	Medium
B4	10-15 min	Slow
B5	16-25 min	Very slow
B6	26+ min	Needs its own head start entirely

Compatibility rule (needs Harris's confirmation): same band, or one band apart, may go in together. Two or more bands apart requires a staggered add, or a different cut size to bring them closer. This is what catches the potato and spinach case: potato at 2cm dice is B4, spinach is B1, three bands apart.

Size and shape scaling

Time is governed by the shortest distance to the centre, not by volume. A 1cm-thick slice cooks in roughly the time of a 1cm dice, however wide the slice. This is why a dense 1cm potato dice and a soft 4cm pepper piece can land in the same band.

Cut vs the reference cut	Multiply time by	Your correction
Half the thickness	x 0.4	
Reference	x 1	
Double the thickness	x 2.5	
Triple the thickness	x 5	

Shape adjustments, applied after the above: strips, julienne and batons take the smallest dimension then x 0.8. Whole and round items take x 1.5 on top of thickness scaling. Shredded or grated is always B1 regardless of density. Sliced discs go by disc thickness only.

Density classes

Class	Character	Examples	Default band at 1cm
D1	Leaf and herb	spinach, rocket, basil, chard leaf	B1
D2	High-water soft	courgette, tomato, mushroom, pepper	B2
D3	Fibrous medium	broccoli, cauliflower, green bean, leek	B3
D4	Dense starchy	potato, carrot, parsnip, celeriac	B4
D5	Hard root, winter squash	swede, turnip, pumpkin	B4-B5

Vegetables

Times are to tender, in a preheated pan or oven at the method's normal temperature. The final column is deliberately blank - write your corrections there.

Ingredient	Class	Reference cut	Saute	Roast 200C	Boll	Steam	Band	Correction
Spinach	D1	whole leaf	1	-	1	2	B1	✓
Rocket	D1	whole leaf	1	-	-	-	B1	✓
Chard leaf	D1	ribbon	2	-	2	3	B1	✓
Chard stem	D3	1cm	6	15	6	8	B3	✓
Kale	D1	torn	4	10	4	5	B2	✓
Cabbage	D3	1cm ribbon	6	20	6	8	B3	✓
Spring onion	D2	1cm	2	-	-	-	B1	✓
Onion, soften	D2	1cm dice	5	25	-	-	B2	✓
Onion, caramelise	D2	1cm dice	35	-	-	-	B6	✓
Shallot	D2	sliced	4	20	-	-	B2	✓
Garlic	D2	minced	1	-	-	-	B1	✓
Leek	D3	1cm slice	6	20	6	7	B3	✓
Mushroom, button	D2	quartered	6	18	-	-	B3	✓
Mushroom, sliced	D2	5mm slice	4	15	-	-	B2	✓
Courgette	D2	1cm dice	5	18	4	5	B2	✓
Aubergine	D2	2cm dice	9	25	-	-	B3	✓
Bell pepper	D2	2cm piece	6	20	-	-	B3	✓
Bell pepper	D2	1cm strip	4	15	-	-	B2	✓
Tomato, fresh	D2	wedge	4	20	-	-	B2	✓
Cherry tomato	D2	whole	5	18	-	-	B2	✓
Asparagus	D3	whole spear	5	12	3	4	B2	✓
Green bean	D3	whole	7	18	5	6	B3	✓
Broccoli floret	D3	3cm floret	7	18	4	5	B3	✓
Cauliflower floret	D3	3cm floret	8	22	6	7	B3	✓
Brussels sprout	D3	halved	9	22	7	8	B3	✓
Fennel	D3	1cm wedge	8	25	7	8	B3	✓
Celery	D3	1cm slice	6	-	6	7	B3	✓
Pak choi	D2	halved	4	-	3	4	B2	✓
Carrot	D4	1cm dice	10	25	8	10	B4	✓
Carrot	D4	5mm slice	6	18	5	6	B3	✓
Potato, waxy	D4	2cm dice	15	30	12	15	B4	✓
Potato, waxy	D4	1cm dice	8	20	7	9	B3	✓
Potato, floury	D4	2cm dice	14	30	15	18	B4	✓
Sweet potato	D4	2cm dice	12	25	10	12	B4	✓
Parsnip	D4	2cm dice	13	28	10	12	B4	✓
Celeriac	D4	2cm dice	14	30	12	14	B4	✓
Beetroot	D4	2cm dice	20	45	25	25	B5	✓
Swede	D5	2cm dice	18	40	18	20	B5	✓

Ingredient	Class	Reference cut	Saute	Roast 200C	Boll	Steam	Band	Correction
Turnip	D5	2cm dice	14	30	12	14	B4	✓
Pumpkin, squash	D5	2cm dice	12	28	10	12	B4	✓
Peas, frozen	D2	whole	2	-	2	3	B1	✓
Sweetcorn kernels	D2	whole	3	15	3	4	B2	✓